INORGANIC CUME

9-28-2012

(1). In fuel cell science, hydrogen gas is a preferred fuel type. Discuss a famous industrial process for making hydrogen gas at an ultra-low cost (using the chemical equation), and then explain why hydrogen is better/worse than a hydrocarbon fuel for operating the fuel-cell. (20')

(2). Over the time, the Pt catalyst in the fuel cell can be gradually deactivated by a toxic impurity in the hydrogen fuel. Discuss details of the chemistry in the catalyst deactivation. (20')

(3). If the Pt has an fcc lattice, draw a unit cell of the crystal, and discuss Miller induces for the **facets** of an octahedron-shaped Pt-nanocrystal and a cube-shaped Pt nanocrystal. (20')

(5). Discuss the lattice similarities/differences between an fcc crystal and a Zn-blende crystal. (20')

(4). The Pt is in fact too expensive for most customers. If you have to find a low-cost candidate to substitute for the Pt, name you best candidate by showing your chemistry reasoning. (20')